

Roguelite Game Maker — User Guide

Senior Software Engineering Project, Winter 2026

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1. Getting Started

The Roguelite Game Maker is a hosted web application, so no installation is required.

To access the editor:

1. Open any modern web browser (Chrome recommended)
2. Navigate to: <http://216.173.116.115/> (Hosted editor)
3. The editor will load automatically and restore your previous session if one exists

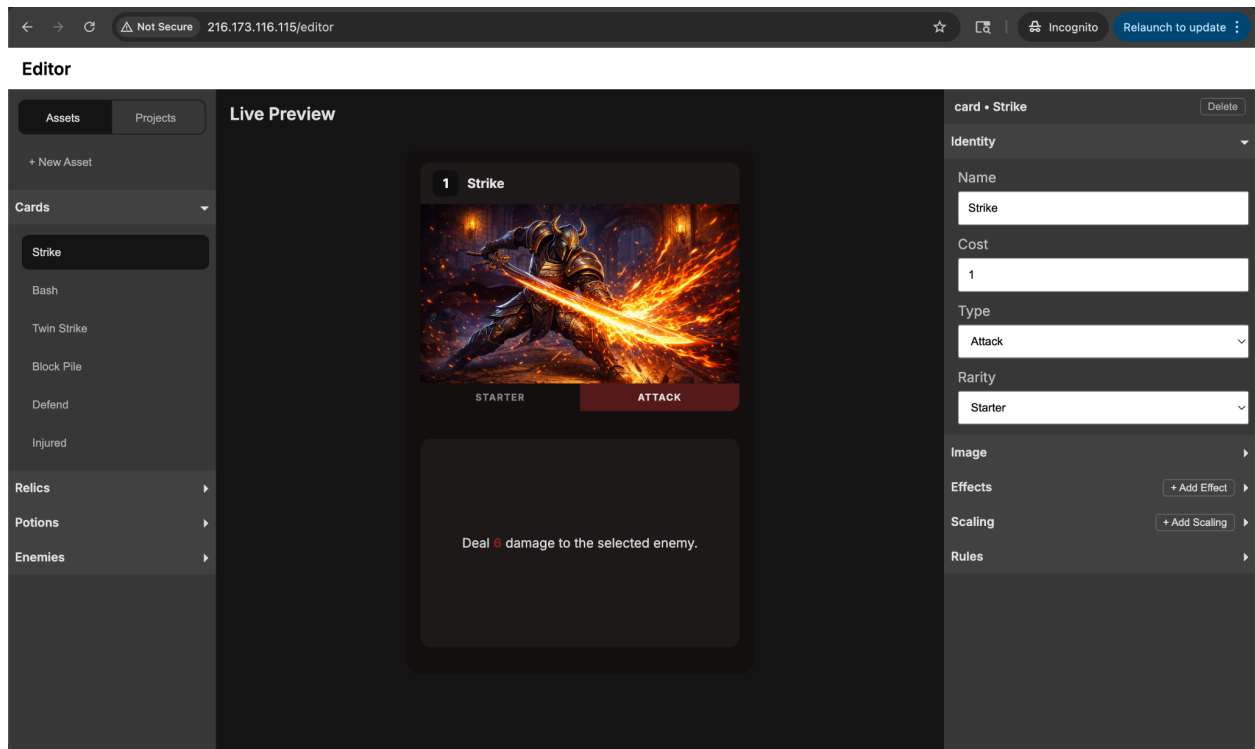


Figure 1. Editor on first load, in Asset Mode with Strike card selected

Your work is saved automatically. Every change you make is synced to the server within one second. You can close the tab to return later, and your project will be exactly as you left it.

2. Editor Layout

The editor has three main components, which make up the full functionality of creating/configuring assets, assigning assets to projects, and running the PyGame or exporting the JSON configuration file.

- **Left Sidebar** - Navigation between assets and projects. Use the toggle at the top to switch between Asset Mode and Project Mode. Click any entity in the list to select it.
- **Center Canvas** - live preview of the selected entity. Updates in real-time as you edit.
- **Right Inspector** - Configuration panel for the selected entity. All editing happens here.

3. Asset Mode (Creating Game Assets)

Asset Mode is where you define the building blocks of your game. There are four asset types: **Cards**, **Relics**, **Potions**, and **Enemies**.

3.1 Creating an Asset

One of the most important features of this application is the ability for users to create their own assets to be used within their games.

To create an asset:

1. In the left sidebar, make sure **Assets** is selected at the top.
2. Select **+ New Asset**, choose asset type, give it a name, and click **Create**.
3. Your new asset should now be the selected asset.
4. Configure your created asset with the inspector and see changes through the live preview.

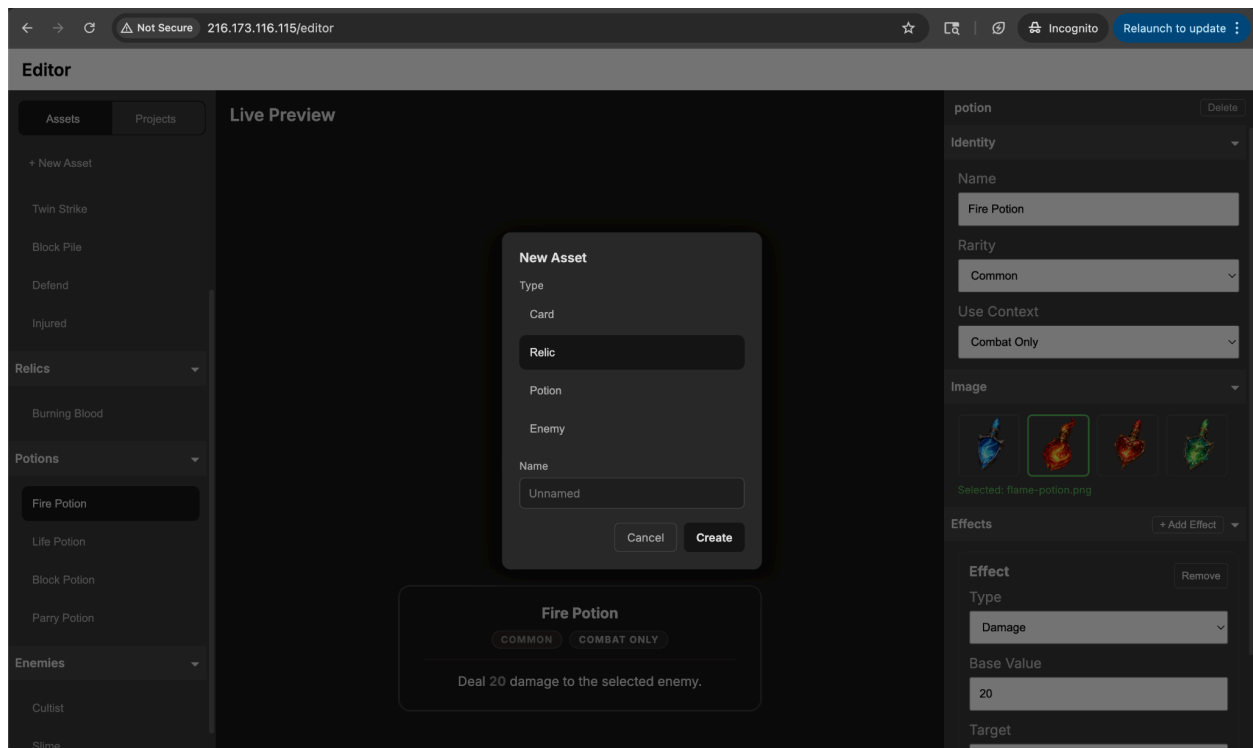


Figure 2. New Asset modal for creating a new asset with a name

3.2 Configuring Assets

Each asset type has the same editing pattern. First select the asset in the sidebar, then use the Inspector to expand and configure each section. Changes appear live in the canvas.

To configure an asset:

1. Make sure the asset you want to edit is selected in the left sidebar.

2. You should now see the content within the preview, and the inspector is updated.
3. The **Identity** section is used to edit things such as name, cost, type, rarity, etc.
4. The **Image** section is used to select the image for the asset.
5. The **Effects** section is used to edit what the asset actually does. Here, you can delete/edit existing effects or add a new effect. Each effect has its own **type** (damage, block, heal, etc.), **base value**, **target**, and **repeat count**.
6. Other sections are **Scaling**, **Triggers**, and **Moves**. These sections tell the asset information about effects, such as if it should scale, when it should trigger, or combine effects into one move.

4. Project Mode (Configuring a Game)

Once assets are created, switch to **Project** Mode using the toggle at the top of the sidebar. This is where you build a complete playable game configuration.

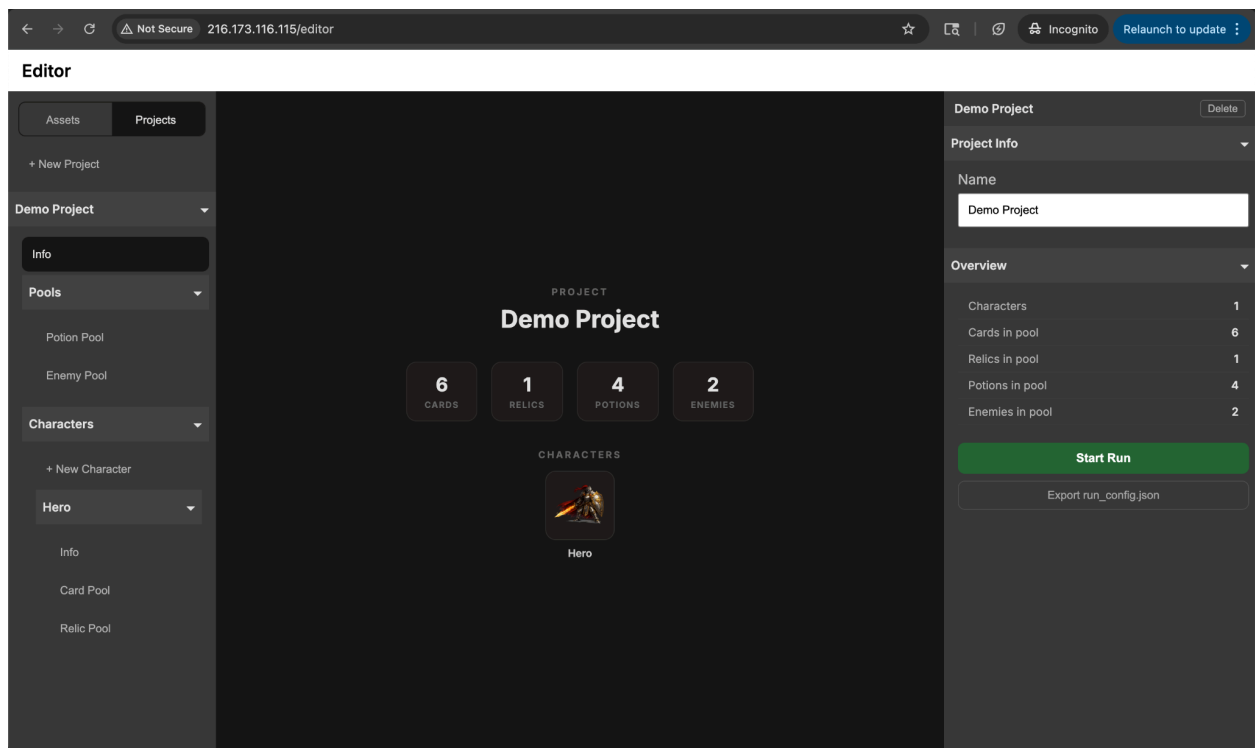


Figure 3. Editor in Project Mode with Demo Project selected

4.1 Creating a Project

One of the most important features of this application is the ability for users to create projects using their own assets.

To create a project:

1. In the left sidebar, make sure **Projects** is selected at the top.
2. Select **+ New Project**, give it a name, and click **Create**.

3. The sidebar expands to show Info, Pools (Potion Pool, Enemy Pool), and Characters.

To create a character:

1. In the left sidebar, make sure **Projects** is selected at the top.
2. Expand the project you would like to make a character for.
3. Expand the **Characters** section and click + **New Character**.
4. Your new character should now be the selected entity.
5. Configure your created character with the inspector and see changes through the live preview.

4.2 Configuring Projects

With a project's **Info** selected, the Inspector shows an overview of the project's current configuration, including character count, pool sizes, and act assignments. Use the sidebar tree to navigate into specific sections.

To configure a project pool:

1. Make sure the project pool you want to edit is selected in the left sidebar.
2. You should now see the content within the preview, and the inspector is updated.
3. Click an asset from the **Inspector** to include/exclude it from the selected project.

To configure a character:

1. Make sure the character you want to edit is selected in the left sidebar.
4. You should now see the content within the preview, and the inspector is updated.
5. Within the **Info** section, choose a starting relic and cards to be included in the starter deck.
6. Within the **Pools** sections, include/exclude relics and cards to be included in the selected characters' reward pools.

5. Running or Exporting

To run or export a project, ensure you are in **Project** mode within the **Info** section of the project you would like to run or export.

- **Start Run** - Launches the pygame viewer locally (requires Django running on your machine)
- **Export run_config.json** - Downloads the game configuration as a JSON file. Use this when running from the hosted editor, by dropping the file into roguelite_map_generator/ and running python main.py

5. Known Alpha Limitations

The following are known limitations as of the alpha release. These will be addressed in future iterations.

- Single shared session per browser; no login or auth system is currently in place

- Asset images are fixed; no custom images are supported
- The start run button is local only; it doesn't work if Django is not running on your local machine
- Game engine in early development; many defined effects are unused in the PyGame viewer